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# Case 2 Risk management report

## Part 1

The Normal Delta approach estimates portfolio risk by linearly approximating the change in option values based on their delta exposures to underlying assets, assuming normally distributed returns. At a 99% confidence level, the total portfolio Value-at-Risk (VaR) using the Asset-Normal Delta approach is CHF 14.3 million, and the Expected Shortfall (ES) is CHF 16.36 million. These means that in the worst 1% of case, losses could exceed CHF 14.3M, with average losses in that tail being even higher. This means the portfolio could lose over CHF 14.3 million in very bad market conditions, and on average even more in the worst 1% of cases. However, combining the positions helps reduce the total risk compared to looking at each position separately.

Using the SMI beta exposure, the portfolio's VaR(99%,1) is CHF 8,414,917.03 and the ES(99%,1) is CHF 10,391,047.30. This means that in the worst 1% of cases, the portfolio could lose over CHF 10.3 million, confirming substantial market sensitivity through the SMI. The SMI mapping approach simplifies risk estimation by mapping all underlying assets to a single market factor using each asset's beta. This assumes that the primary source of risk is the overall market (SMI), not the individual asset movement.

The Historical Simulation approach yields a VaR(99%,1) of CHF 15,539,143.46 and an ES(99%,1) of CHF 19,296,856.75. These are higher than those obtained via the beta-based SMI exposure approach (VaR ≈ CHF 8.41M, ES ≈ CHF 10.39M), but slightly lower than the Delta-Normal results (VaR ≈ CHF 14.3M, ES ≈ CHF 16.36M). This suggests that historical simulation captures more extreme outcomes than the beta-SMI mapping but is more conservative than the linear approximation of the Delta-Normal method.

## Part 2

Using today's term structure, the present values of the three bond positions were calculated as CHF 96.20, CHF 103.45, and CHF 91.62 respectively. Multiplying these by the number of bonds (accounting for long/short positions), the total portfolio value as of April 25, 2025, is CHF 403,699,002.73. This represents the current market value of the fixed-income portfolio based on discounted future cash flows.

Using the cash-flow mapping and 250-day yield change VCV matrix, the 1-day Asset-Normal VaR at 99% confidence is CHF 1,969,010.24, and the ES is CHF 2,252,108.28.

This result reflects the risk of the fixed income portfolio based on how interest rate changes affect future cash flows. The relatively modest values (compared to previous equity portfolio risks) indicate that the bond portfolio is less volatile under normal market conditions.

the Historical Simulation Approach yields a 1-day 99% VaR of $2,112,647.39 and an Expected Shortfall of $2,491,936.89. These values are slightly lower than those obtained with the Asset-Normal approach, suggesting that the simulation reflects a somewhat less extreme tail risk under recent historical yield movements. This might indicate more stable or less volatile recent conditions in the fixed-income market.

## Part 3

For regulatory reporting, the total 1-day risk at the 99% confidence level for the entire fund (equity + fixed income) is: VaR(99%,1): CHF 21,882,205.42 and ES(99%,1): CHF 28,850,817.85. These figures combine the risks from both portfolios, accounting for a low positive correlation (0.13) between them, which modestly reduces the overall risk. The fixed income portfolio was converted from USD to CHF to ensure consistency in reporting.